ORIGINAL ARTICLE

Mitochondrial Donation and Preimplantation Genetic Testing for mtDNA Disease

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ABSTRACT

BACKGROUND

Children born to women who carry pathogenic variants in mitochondrial DNA (mtDNA) are at risk for a range of clinical syndromes collectively known as mtDNA disease. Mitochondrial donation by pronuclear transfer involves transplantation of nuclear genome from a fertilized egg from the affected woman to an enucleated fertilized egg donated by an unaffected woman. Thus, pronuclear transfer offers affected women the potential to have a genetically related child with a reduced risk of mtDNA disease.

METHODS

We offered mitochondrial donation (by pronuclear transfer) or preimplantation genetic testing (PGT) to a series of women with pathogenic mtDNA variants who sought to reduce the transmission of these variants to their children. Patients with heteroplasmy (variants present in a proportion of copies of mtDNA) were offered PGT, and patients with homoplasmy (variants present in all copies of mtDNA) or elevated heteroplasmy were offered pronuclear transfer.

RESULTS

Clinical pregnancies were confirmed in 8 of 22 patients (36%) and 16 of 39 patients (41%) who underwent an intracytoplasmic sperm injection procedure for pronuclear transfer or for PGT, respectively. Pronuclear transfer resulted in 8 live births and 1 ongoing pregnancy. PGT resulted in 18 live births. Heteroplasmy levels in the blood of the 8 infants whose mothers underwent pronuclear transfer ranged from undetectable to 16%. Levels of the maternal pathogenic mtDNA variant were 95 to 100% lower in 6 newborns and 77 to 88% lower in 2 newborns than in the corresponding enucleated zygotes. Heteroplasmy levels were known for 10 of the 18 infants whose mothers underwent PGT and ranged from undetectable to 7%.

CONCLUSIONS

We found that mitochondrial donation through pronuclear transfer was compatible with human embryo viability. An integrated program involving pronuclear transfer and PGT was effective in reducing the transmission of homoplasmic and heteroplasmic pathogenic mtDNA variants. (Funded by NHS England and others.)

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The mitochondrial genome is inherited maternally, and children born to women who carry pathogenic mitochondrial DNA (mtDNA) variants are at risk for life-limiting mtDNA disease.[1,2] Disorders resulting from pathogenic mtDNA variants have an estimated prevalence of 1 in 5000 births.[2] Pathogenic variants can be either homoplasmic (present in all copies of mtDNA) or heteroplasmic (present in a proportion of copies of mtDNA). Homoplasmic variants are transmitted in full to all children, but the penetrance of homoplasmic variants varies.[3] Transmission of heteroplasmic variants is subject to a genetic bottleneck. This mtDNA genetic bottleneck gives rise to a random shift in mtDNA heteroplasmy, which results in widely varying levels of heteroplasmy among oocytes from the same woman,[3] making the risk of the development of severe disease in a child difficult to predict.[4]

In the absence of curative treatments, advances in assisted reproductive technology offer the possibility to reduce transmission of pathogenic mtDNA variants.[5-7] Preimplantation genetic testing (PGT), which is used widely for detecting nuclear genetic defects,[6,8] can be used to reduce the risk of mtDNA disease by identifying embryos with low levels of the maternal mtDNA variant.[9] In the event that an embryo with low levels of heteroplasmy is not available, mitochondrial donation (or mitochondrial replacement) offers an alternative means of potentially reducing the risk of mtDNA disease; this process involves the microsurgical transplantation of the nuclear genome from an egg of a woman carrying the mtDNA variant to an enucleated egg from an unaffected woman.[5,7,10]

Mitochondrial donation can be performed before fertilization during metaphase II arrest (maternal-spindle transfer) or after fertilization when the maternal and paternal haploid genomes are contained in pronuclei (pronuclear transfer).[10] Both approaches can result in embryos with heteroplasmy for maternal mtDNA owing to cotransfer (carryover) of some mitochondria surrounding the transplanted pronuclei.[11-13] For reasons that remain unclear, the small amount of maternal mtDNA increases to homoplasmic levels in approximately 20% of embryonic stem-cell lines derived from embryos obtained after a mitochondrial donation procedure.[11,14,15] Moreover, an elevated level of heteroplasmy (40 to 60%) for maternal mtDNA has been reported in one of six babies born after maternal-spindle transfer for infertility treatment.[16] These findings evoke questions about whether mitochondrial donation can reliably prevent transmission of pathogenic mtDNA variants in all cases, particularly in cases of homoplasmy.[17] Here we describe the clinical application of a pronuclear-transfer procedure previously established for human zygotes.[11] Pronuclear transfer and PGT were offered in an integrated program to reduce transmission of a range of heteroplasmic and homoplasmic pathogenic variants. Maternal and child health outcomes are described by McFarland et al. in an accompanying article in this issue of the *Journal*.[18]

Methods

Study Oversight

Treatment procedures for assisted reproductive technology including pronuclear transfer were conducted in accordance with Human Fertilisation and Embryology Authority (HFEA) center license 0017. All outcomes were reported to the HFEA. Clinical pronuclear-transfer procedures were performed by an HFEA-licensed pronuclear-transfer practitioner. Patient-specific approval for pronuclear-transfer treatment was obtained from the HFEA Statutory Approvals Committee. Research was approved by the Newcastle and North Tyneside Research Ethics Committee and was licensed by the HFEA. The authors vouch for the completeness and accuracy of the data.

Procedures

Patients with heteroplasmy were offered PGT, and patients with homoplasmy or elevated heteroplasmy were offered pronuclear transfer. Patients undergoing PGT and those undergoing pronuclear transfer underwent health assessments and standard fertility assessments before undergoing ovarian stimulation and oocyte retrieval. In addition, egg donor mtDNA was sequenced to screen for pathogenic mtDNA variants. All the patients and donors were offered counseling before entering the program and throughout the treatment pathway (see the Supplementary Methods section of the Supplementary Appendix, available with the full text of this article at NEJM.org).

On the basis of evidence from preclinical research,[11] the standard practice is to vitrify the eggs of patients undergoing pronuclear transfer

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and store them until fresh donor eggs become available. All eggs were fertilized by intracytoplasmic sperm injection. Pronuclear transfer was performed at 8 to 13 hours after intracytoplasmic sperm injection, and intact zygotes were cultured for 5 to 6 days. For the PGT treatment, a single blastomere was removed on day 3 after intracytoplasmic sperm injection for quantitative analysis of mtDNA heteroplasmy,[19] and embryos were cultured for an additional 2 to 3 days. Embryos of suitable quality were used for fresh intrauterine transfer or vitrified for a future frozen-embryo transfer. In the PGT group, embryos with less than 30% heteroplasmy were preferentially used for transfer or vitrification (Fig. 1). Infant peripheral blood was used to assess heteroplasmy for the maternal pathogenic variant (see the Supplementary Methods). Data were analyzed with a chi-square test, Fisher's exact test, unpaired t-test, and Tukey's multiple comparison test with the use of GraphPad Prism, version 9.0.1.

RESULTS

PATIENTS

A total of 32 patients were granted case-specific HFEA approval for pronuclear transfer. Here we describe the treatment outcomes of women who

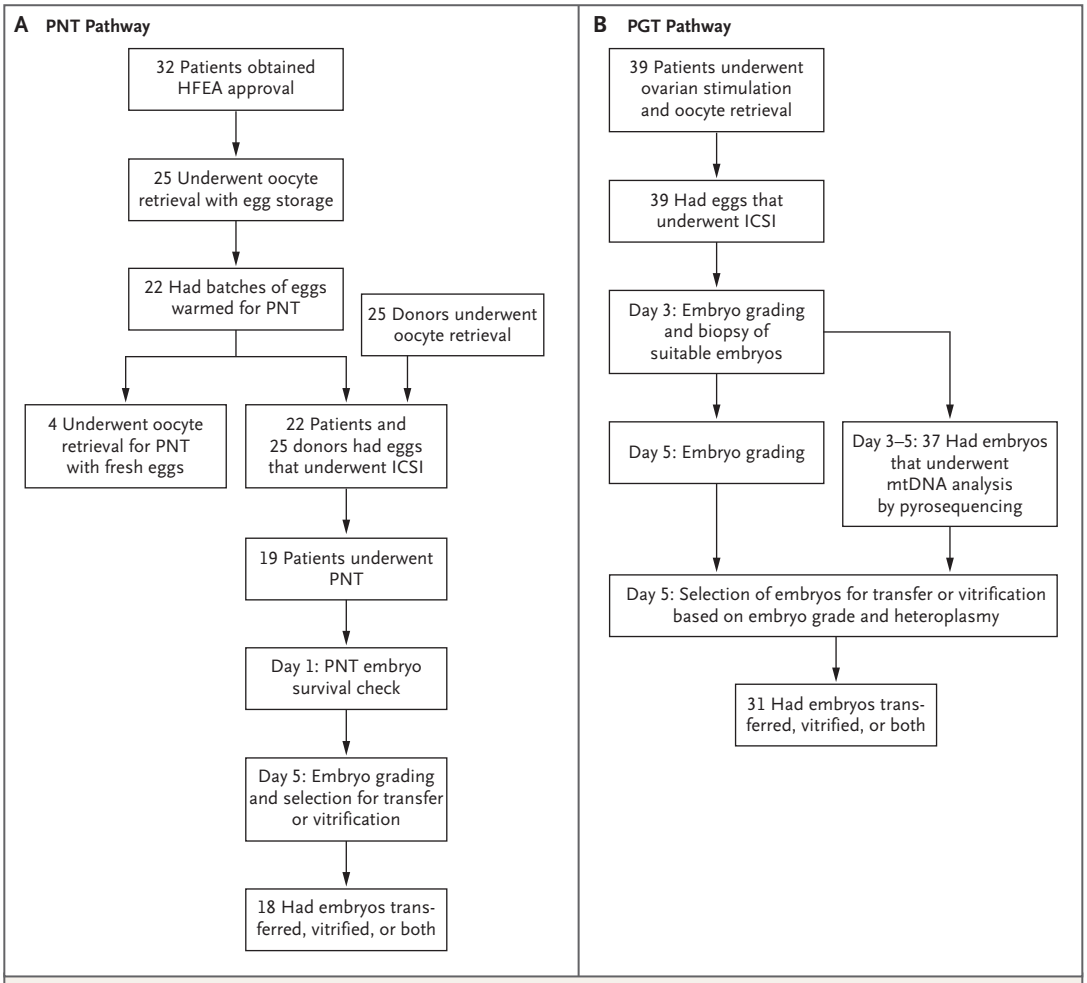


Figure 1. Pronuclear Transfer and Preimplantation Genetic Testing Treatment Pathways.

Panel A shows the pronuclear-transfer (PNT) pathway and the progression of the 32 patients approved for pronuclear transfer through the stages of egg storage, pronuclear transfer, and embryo transfer. Panel B shows the preimplantation genetic testing (PGT) pathway and the progression of the 39 patients through the stages of intracytoplasmic sperm injection (ICSI), embryo biopsy, and embryo transfer. HFEA denotes Human Fertilisation and Embryology Authority, and mtDNA mitochondrial DNA.

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underwent an oocyte retrieval procedure for pronuclear transfer (25 patients) or who underwent PGT (39 patients) to reduce transmission of pathogenic mtDNA variants (Table S1 in the Supplementary Appendix and Fig. 1). Most patients had White European ancestry (Table S2).

Mitochondrial DNA encodes 13 respiratory chain complex subunits, 22 transfer RNAs (mt-tRNAs), and 2 ribosomal RNAs.[1] Among patients who were included in one of the two treatment groups, pathogenic variants were present in either protein-coding (in 26 patients) or mt-tRNA genes (in 33 patients) and were homoplasmic in 12 patients and heteroplasmic in 47 patients. One patient had a large-scale mtDNA deletion (m.7529_14025del in accordance with revised Cambridge reference sequence number NC_012920.1). PGT was offered in most cases of heteroplasmy (39 patients), and pronuclear transfer was offered in cases of homoplasmy (12 patients) and in some cases of high heteroplasmy (13 patients) (Fig. S1). The latter included 3 patients who were offered pronuclear transfer after previously undergoing at least one cycle of PGT as part of this study.

Pathogenic variants in protein-coding mtDNA sequences were present in *MT-ND* genes (encoding complex I subunits) in 17 patients or in *MT-ATP6* (encoding a complex V subunit) in 9 patients. Variants in *MT-ND* included Leber's hereditary optic neuropathy (LHON) variants (in 11 patients) and non-LHON variants (in 6 patients). LHON variants are typically homoplasmic, and LHON carriers accounted for the largest cohort (44%) of patients undergoing pronuclear transfer. All other protein-coding variants were heteroplasmic; women carrying these variants underwent PGT. Similarly, most patients carrying mt-tRNA variants (70%) underwent PGT. The m.3243A→G *MT-TL1* variant was the most common, and 17 of the 21 women who carried this variant underwent PGT (1 of these 21 women was included in both groups). Pronuclear transfer was offered to 5 patients with elevated heteroplasmy for the m.3243A→G *MT-TL1* variant and to 4 patients with elevated heteroplasmy for the m.8344A→G *MT-TK* variant, including 3 patients who had previously undergone PGT as part of this study. Five patients who underwent pronuclear transfer had homoplasmy (4 patients) or elevated heteroplasmy (1 patient) for mt-tRNA variants referred to here as rare mt-tRNA variants.

OOCYTE RETRIEVAL AND FERTILIZATION

The age of the patients was similar in the PGT group (median, 34 years; range, 22 to 40) and the pronuclear-transfer group (median, 34 years; range, 25 to 40) (Fig. S2). Both patient cohorts underwent one to three ovarian-stimulation and oocyte-retrieval procedures, which resulted in 659 oocytes (mean [±SD] number of oocytes per retrieval, 10.6±6.7) in the pronuclear-transfer group and 586 oocytes (mean number per retrieval, 10.3±5.5) in the PGT group (Fig. S3 and Fig. 2A). The number of oocytes retrieved was similar in the PGT group, the pronuclear-transfer group, and an age-matched control group. A total of 25 egg donors (21 to 37 years of age) underwent 38 ovarian-stimulation and oocyte-retrieval procedures, which resulted in 736 oocytes (mean, 19.4±8.7). The percentage of oocytes that were mature (eggs) at the time of retrieval was equivalent in the two patient cohorts (79.4% in the PGT group and 78.8% in the pronuclear-transfer group; $P=0.80$ by chi-square test) and in the egg-donor group (80.2%; $P=0.81$ by chi-square test for the comparison of the egg-donor group with the two patient groups).

A total of 487 eggs obtained from the 25 patients in the pronuclear-transfer group were vitrified and placed in cryostorage (Fig. 2B). Of these 487 vitrified eggs, 327 from 22 patients were removed from storage in batches of 3 to 14 eggs on the day of the donor oocyte retrieval; 325 (99.4%) survived the vitrification and warming procedures. A total of 32 fresh eggs and 16 vitrified eggs were used for pronuclear transfer in 4 women (Fig. 1A). Fresh eggs were used because of concerns about fertilization (1 woman) or endometrial thickness (1 woman) in a previous cycle or to expedite treatment (2 women).

Intracytoplasmic sperm injection was performed in 357 eggs obtained from the patients undergoing pronuclear transfer with the use of sperm from the patients' partner (21 patients) or from a donor (1 patient). A total of 572 donor eggs and corresponding patient eggs were injected with sperm from the same source. However, in three cases, the use of donor sperm (i.e., from a different source than used for the patient eggs) to fertilize donor eggs was mandated because the egg donor was genetically related to the patients' male partner.

Normal fertilization, evidenced by the presence

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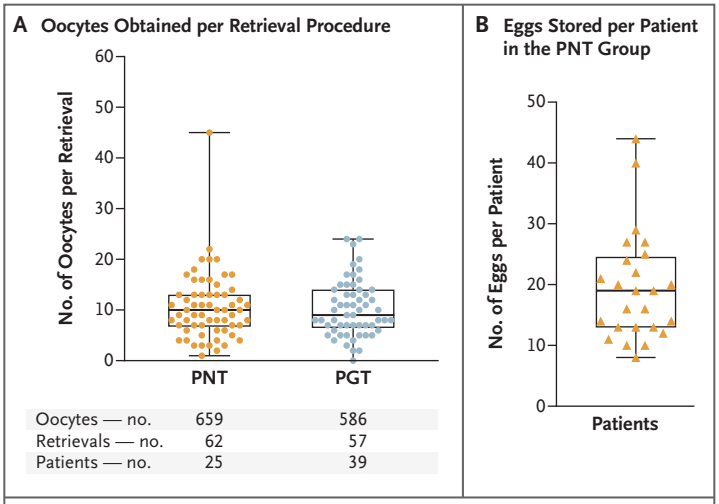
of two pronuclei, was observed in 48.5% of eggs from the pronuclear-transfer group, as compared with 63.8% of donor eggs (P<0.001 by chi-square test) and 66.5% of eggs from the PGT group (P<0.001 by chi-square test) (Fig. 2C). Fertilization of eggs from the pronuclear-transfer group may have been affected by vitrification (Fig. S4). However, we observed variation in the success of fertilization according to the pathogenic variant. Three patients had no fertilized eggs available for pronuclear transfer (Table S3).

PRONUCLEAR TRANSFER, EMBRYO BIOPSY, AND DEVELOPMENT

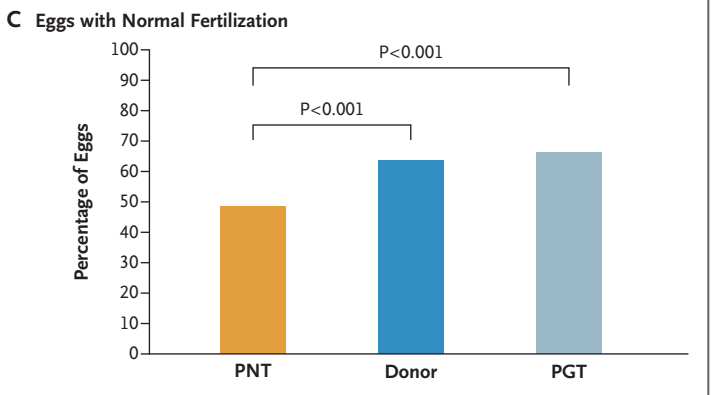
Pronuclear transfer was performed 8 to 13 hours after intracytoplasmic sperm injection[11] between 160 pairs of normally fertilized zygotes from 19 patients in the pronuclear-transfer group and 24 egg donors. Pronuclei were removed from the patient zygote separately and placed together under the zona of the enucleated donor egg after brief exposure to a fusion agent (hemagglutinating virus of Japan envelope) (Fig. 3A and B and Video 1). Semiquantitative scores were assigned to estimate the volume of cytoplasm carried over in patient karyoplasts (scores ranged from 1 to 4, with higher scores indicating a higher volume of cytoplasm) and to provide an indicator of mitochondrial donation–induced heteroplasmy. A score was also assigned for the volume of cytoplasmic leakage from donor eggs (Fig. S5). Replacement of the pronuclei from the donor's zygote with the pronuclei from the patient's zygote was successful in 127 of 160 attempts (79.4%); of the 127 embryos resulting from these successful attempts, 122 (96.1%) were intact the next day (day 1). The number of intact zygotes per each of the 33 pronuclear-transfer procedures ranged from 0 to 7 (Fig. S6).

Embryos from 37 of the 39 patients (95%) in the PGT group were assessed on day 3 after intracytoplasmic sperm injection, and embryos with six or more cells (262 of the 309 zygotes [84.8%]) had a single blastomere removed[19] for mtDNA analysis. The percentage of embryos that developed to the blastocyst stage was similar in the pronuclear-transfer group and the PGT group on day 5 (45.1% and 46.3%, respectively; P=0.82 by chi-square test) and day 6 (46.7% and 50.8%, respectively; P=0.44 by chi-square test) (Fig. 3C). The combined percentage of top-quality and good-quality blastocysts[20] was lower with PGT than with pronuclear transfer (42.0% vs. 63.2% on day 6; P=0.006 by chi-square test) (Fig. 3D and 3E).

Our current practice is to preferentially exclude blastocysts with more than 30% heteroplasmy from use in fertility treatment. However,



	PNT	PGT
Oocytes — no.	659	586
Retrievals — no.	62	57
Patients — no.	25	39



	PNT	Donor	PGT
Eggs — no.	357	572	465
ICSI procedures — no.	38	38	56
Patients/donors — no.	22	25	39

Figure 2. Oocyte Retrieval and Fertilization Outcomes.

Panel A shows the number of oocytes obtained per retrieval procedure among patients who underwent pronuclear transfer (PNT) and those who underwent PGT. Dots denote the retrieval procedures. Panel B shows the number of eggs stored per patient who underwent pronuclear transfer. Triangles denote patients. In Panels A and B, the horizontal line within each box indicates the median, the top and bottom of the box indicate the interquartile range, and the whiskers indicate the minimum and maximum values. Panel C shows the percentage of eggs with normal fertilization among patients in the pronuclear-transfer group as compared with those in the PGT group (P<0.001 by chi-square test) and egg donors (P<0.001 by chi-square test).

A video showing the transfer of pronuclei is available at NEJM.org

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decision making varies with respect to selection or exclusion of blastocysts depending on the variant and family history. In this series, 42.4% of top-quality and good-quality PGT blastocysts had more than 30% heteroplasmy and were therefore not used for treatment. Thus, the best-quality PGT embryos could not be used for 25% of fresh-embryo transfers.

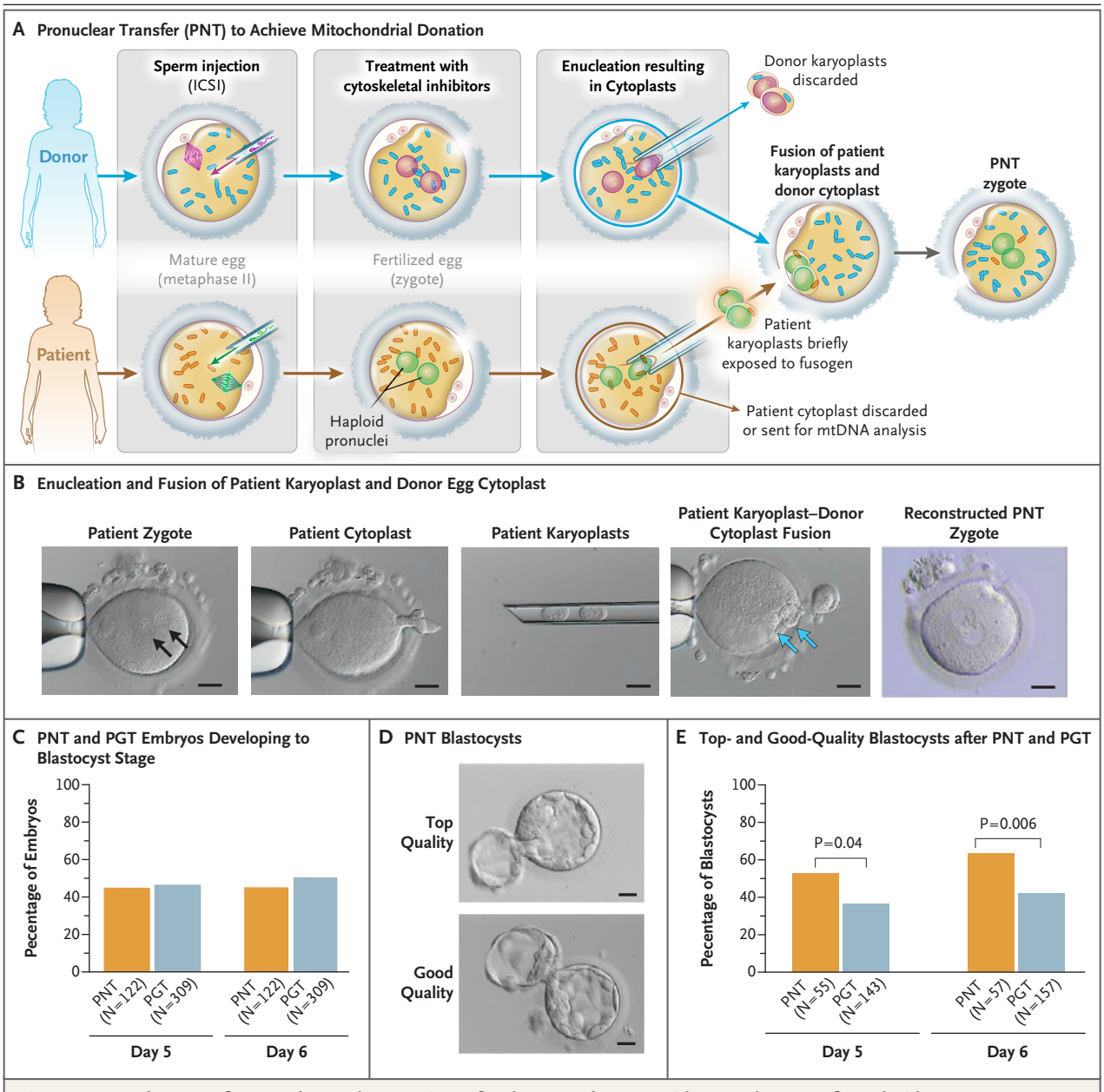


Figure 3. Pronuclear-Transfer Procedure and Comparison of Embryo Development with Pronuclear Transfer and with PGT.

Panel A shows the pronuclear-transfer procedure. The images in Panel B show the enucleation of patient karyoplasts and the fusion of patient karyoplasts with the donor egg cytoplast (scale bar, 20 μm). Arrows in the first image indicate the pronuclei, and arrows in the fourth image indicate the karyoplasts. Panel C shows a comparison of the percentage of embryos with blastocyst formation on days 5 and 6 after intracytoplasmic sperm injection in the pronuclear-transfer group and in the PGT group. The images in panel D show examples of top-quality and good-quality pronuclear-transfer blastocysts (scale bar, 20 μm). Panel E shows the percentage of blastocysts that were top quality or good quality after pronuclear transfer and after PGT on days 5 and 6. The percentage was higher in the pronuclear-transfer group than the PGT group on both day 5 (P=0.04 by chi-square test) and day 6 (P=0.006 by chi-square test).

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EMBRYO TRANSFER AND TREATMENT OUTCOMES

A total of 73.7% of intracytoplasmic sperm injections in the pronuclear-transfer group and 66.7% of those in the PGT group resulted in the transfer of fresh or frozen embryos (Fig. S7). In five cases, all suitable embryos were vitrified owing to concerns regarding the risk of ovarian hyperstimulation or poor endometrial response after hormonal priming for embryo transfer. Of the 22 patients in the pronuclear-transfer group and the 39 in the PGT group who underwent intracytoplasmic sperm injection, 82% and 79%, respectively, had fresh or frozen embryos (or both) available for transfer (Fig. 4A). The predominant reasons for having no embryos available for transfer were failure of fertilization in the pronuclear-transfer group (3 patients) and failure to obtain embryos with a heteroplasmy level of less than 30% in the PGT group (5 patients). A single embryo was transferred during all 40 embryo-transfer procedures in the pronuclear-transfer group. In the PGT group, a single embryo was transferred in 42 of the 51 embryo-transfer procedures; 2 embryos were transferred during 9 procedures owing to concerns about the woman's age (1 procedure) or embryo quality (8 procedures).

Among the patients who underwent an intracytoplasmic sperm injection, the overall incidence of clinical pregnancy (confirmed by ultrasonography at 7 weeks of gestation) was similar in the two groups (8 of 22 patients [36%] in the pronuclear-transfer group and 16 of 39 [41%] in the PGT group; P=0.72 by chi-square test) (Fig. 4B). However, the percentage of fresh and frozen embryo transfers that resulted in a clinical pregnancy was lower in the pronuclear-transfer group than in the PGT group (20% vs. 39%; P=0.05 by chi-square test) (Fig. 4C). In addition, the incidence of biochemical pregnancy after a positive pregnancy test was higher in the pronuclear-transfer group than the PGT group (P=0.04 by Fisher's exact test) (Fig. 4D), particularly when fresh embryos had been transferred (P=0.02 by Fisher's exact test). A total of 3 of 6 patients in the pronuclear-transfer group who had a biochemical pregnancy had a viable clinical pregnancy after a subsequent embryo transfer; in 2 of these 3 patients, the earlier biochemical pregnancy and the later clinical pregnancy involved eggs from the same donor. Thus, the reduced viability of pronuclear-transfer embryos is not readily explained by intrinsic maternal factors or by an effect of potentially unfavorable combinations of nuclear and mitochondrial factors. Of the 20 confirmed clinical pregnancies in the PGT group, 3 (15%) ended in miscarriage. None of the 8 clinical pregnancies

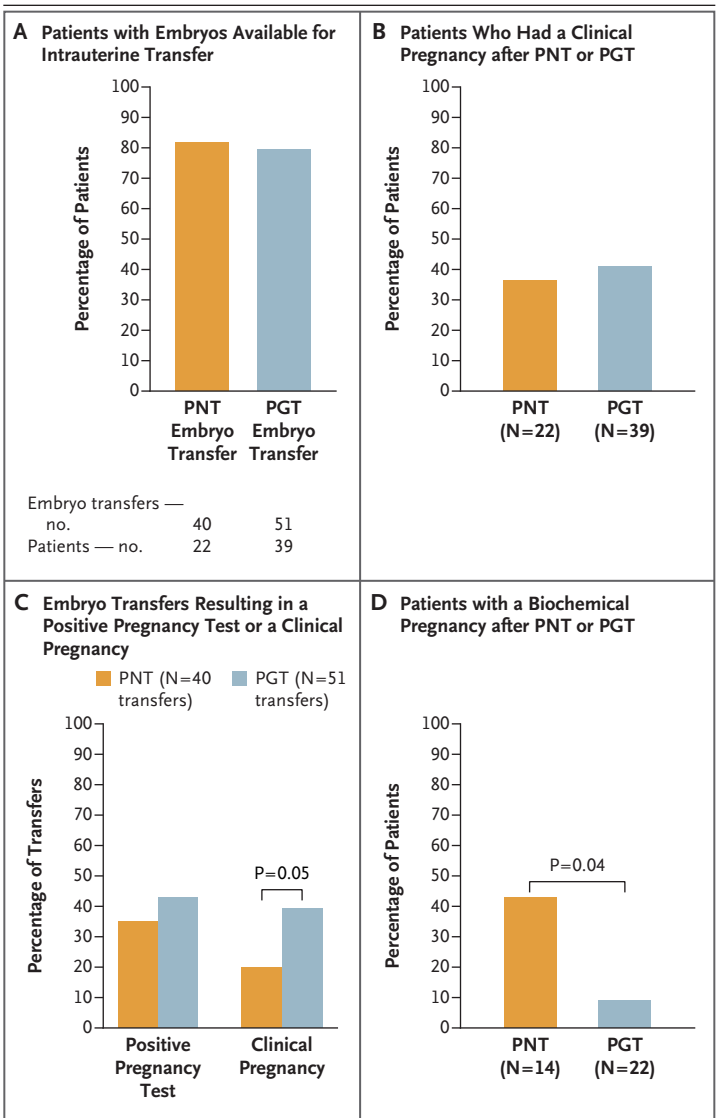


Figure 4. Clinical Outcomes after Pronuclear Transfer and PGT.

Panel A shows the percentage of patients who had embryos available for intrauterine transfer after intracytoplasmic sperm injection for pronuclear transfer or PGT. Panel B shows the overall percentage of patients who had a clinical pregnancy after pronuclear transfer or PGT. Panel C shows the percentage of embryo transfers (fresh and frozen) that resulted in a positive pregnancy test and those that resulted in a clinical pregnancy after pronuclear transfer or PGT. Patients who underwent pronuclear transfer had a lower incidence of clinical pregnancy than those who underwent PGT (P=0.05 by chi-square test). Panel D shows the percentage of patients with a biochemical pregnancy after pronuclear transfer or PGT. Patients who underwent pronuclear transfer had a higher incidence of biochemical pregnancy than those who underwent PGT (P=0.04 by Fisher's exact test).

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in the pronuclear-transfer group ended in miscarriage (Table S4).

Of the 8 patients who had a clinical pregnancy after pronuclear transfer, 5 carried primary LHON variants and 3 carried rare mt-tRNA variants. None of the three patients in the pronuclear-transfer group who carried the m.3243A→G variant or of the 4 patients who carried the m.8344A→G variant had a clinical pregnancy. Of the 16 patients who had at least one clinical pregnancy after PGT, half carried protein-coding variants (4 carried *MT-ATP6* gene variants and 4 carried non-LHON *MT-ND* gene variants) and half carried mt-tRNA variants (6 carried m.3243A→G and 2 carried m.8344A→G). These findings show the success of both approaches in achieving viable pregnancies in women with a wide range of mtDNA variants.

LIVE BIRTHS

Of the 21 patients who had a live birth after PGT or pronuclear transfer, 6 had previously given birth to an affected child. In this study, pronuclear transfer resulted in 8 live births (4 female and 4 male infants, including monozygotic twins) and 1 ongoing pregnancy. PGT resulted in 18 live births (9 female and 9 male infants) (Fig. S8). One patient in the PGT group gave birth to a set of dizygotic twins, and 3 patients had 2 singleton live births. There was no significant difference in the gestational age of babies at birth between the pronuclear-transfer group and the PGT group ($P=0.41$ by unpaired t-test). Similarly, an analysis of birth weights according to gestational age showed no significant difference between the two groups ($P=1.00$ by Fisher's exact test). There were no reports of congenital abnormalities at birth. However, a cardiac anomaly was detected in 1 infant born after PGT to a mother with the m.3243A→G variant and in 1 born after pronuclear transfer to a mother with the m.4300G→A variant. In both cases, infant heteroplasmy for the maternal pathogenic variant was 5% or less.

HETEROPLASMY AFTER PRONUCLEAR TRANSFER AND PGT

Regulations in the United Kingdom do not permit assay of heteroplasmy in embryos obtained by pronuclear transfer (HFEA Code of Practice section 33, edition 9.4, October 2023). Moreover, none of the patients in the pronuclear-transfer group included in this series opted for prenatal diagnosis. In five of eight newborns, levels of heteroplasmy for the maternal pathogenic variant were undetectable (<3%) as measured with quantitative pyrosequencing of blood-spot tests (Guthrie card). Next-generation sequencing of DNA isolated from blood spots from three of these five newborns showed the level of heteroplasmy ranging from 0.06 to 0.17%. Three newborns had heteroplasmy levels of 5%, 12%, and 16% (Table 1). Next-generation sequencing of DNA from 62 embryos not suitable for transfer or freezing revealed consistently low heteroplasmy levels (ranging from undetectable to 3.9%; mean, 1.18±0.90), regardless of whether the corresponding enucleated zygote from the patient was homoplasmic or heteroplasmic (Fig. S9). Overall, levels of maternal pathogenic mtDNA variant were 95 to 100% lower in six newborns and 77 to 88% lower in two others than in the corresponding enucleated zygotes.

Analysis of variant heteroplasmy in arrested pronuclear-transfer embryos according to the scores assigned for carryover of maternal cytoplasm in karyoplasts during pronuclear transfer showed a correlation between carryover scores and embryo heteroplasmy ($P<0.01$ by Tukey's multiple comparison test). Carryover scores of 1 and 2 were assigned to embryos from pronuclear-transfer procedures that resulted in 12% and 16% neonatal heteroplasmy, respectively. These scores are similar to the scores assigned to embryos from pronuclear-transfer procedures that resulted in less than 1% heteroplasmy (as detected by next-generation sequencing) in infant blood (Table 1). Moreover, heteroplasmy levels in 24 embryos with carryover scores of 1 and in 24 embryos with scores of 2 ranged from 0 to 1.6% (mean, 0.74±0.48) and from 0 to 2.8% (mean, 0.99±0.69), respectively. Thus, it is unlikely that increased carryover of maternal cytoplasm accounts for the relatively high neonatal heteroplasmy levels (12% and 16%) observed in two cases of pronuclear transfer.

Patients who undergo PGT are encouraged to have heteroplasmy levels measured in their child, but not all patients pursue this option. In the current series, heteroplasmy data were available for 10 of 18 children born after PGT. In eight cases involving the transfer of a single embryo, the pathogenic variant was undetectable in the test blastomere and in infant blood (Table 1). The transfer of two embryos, one with undetect-

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Table 1. Maternal Pathogenic Variant Heteroplasmy in Infants.*

Assisted Reproductive Procedure and Maternal Variant	Reproductive History	Percent Heteroplasmy		Pronuclear-Transfer Procedure Score		Percent Heteroplasmy in Neonatal Blood		Infant Sex
		Maternal Urine	Enucleated Egg or Test Blastomere†	Cytoplasm Carryover‡	Cytoplasm Leakage§	Pyrosequencing	NGS	
Pronuclear transfer								
m.4300A→G	No pregnancies or births	Homoplasmic	—	2	2	5	—	Female
m.4300A→G	Live birth of affected child (deceased)	Homoplasmic	—	2	2	Not detected	0.06	Male
m.3260A→G	No pregnancies or births	79	68	2	1	16	—	Male
m.3460G→A	No pregnancies or births	89	99	2	4	Not detected	0.17	Female
m.11778G→A	No pregnancies or births	82	64	1	0	Not detected	0.09	Female
m.11778G→A	No pregnancies or births	Homoplasmic	—	3	1			
Identical twin 1						Not detected	—	Male
Identical twin 2						Not detected	—	Male
m.11778G→A	Live birth	Homoplasmic¶	99.9	1	3	12	—	Female
PGT								
m.10158T→C	Live birth of affected child (deceased)	33						
Child 1			Not detected			Not detected		Male
Child 2			Not detected			Not detected		Female
m.13094T→C	Live birth of affected child (deceased)	45	Not detected			Not detected		Male
m.3243A→G	No pregnancies or births	23	Not detected			Not detected		Female
m.3243A→G	No pregnancies or births	23	Not detected			Not detected		Male
m.3243A→G	No pregnancies or births	61	9 and 24‖					
Twin 1						Not tested		Female
Twin 2						Not detected		Male
m.3688G→A	Live birth of affected child (deceased)	50	Not detected and 5**			7		Female
m.8993T→G	Live birth of affected child (deceased)	Not detected						
Child 1			Not detected			Not detected		Female
Child 2			Not detected			Not detected		Male
m.8993T→C	Live birth of affected child	37	Not detected			Not detected		Female

* Data are shown for 8 infants born to date after pronuclear transfer and 11 infants born after preimplantation genetic testing (PGT). Mitochondrial DNA analysis could not be carried out for 7 children born after PGT. Mitochondrial DNA variant positions are in accordance with revised Cambridge reference sequence number NC_012920.1. NGS denotes next-generation sequencing. Not detected indicates levels were below 3% and were therefore not reliably detected by pyrosequencing.
† Heteroplasmy levels were measured in enucleated eggs in the pronuclear-transfer group and in test blastomeres in the PGT group.
‡ Scores range from 1 to 4, with higher scores indicating a higher volume of maternal cytoplasm carried over in patient karyoplasts.
§ Scores range from 1 to 4, with higher scores indicating a higher volume of cytoplasm leakage from the enucleated donor egg.
¶ Maternal blood showed 94% heteroplasmy.
‖ Which embryo resulted in the birth of the live male infant and which in the birth of the live female infant is unknown.
** Two embryos were transferred, and which embryo resulted in a live birth is unknown.

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able heteroplasmy and one with 5% heteroplasmy in the test blastomeres, resulted in a singleton pregnancy and the birth of an infant with 7% heteroplasmy. It was not possible to determine which embryo resulted in a viable pregnancy. Similarly, transfer of two embryos, one with a test blastomere level of 9% and one with a test blastomere level of 24%, resulted in the birth of twins. One infant had an undetectable heteroplasmy level and the other was not tested (Table 1).

The decision of whether to offer pronuclear transfer or PGT was based on the probability of obtaining embryos with less than 30% heteroplasmy. Eleven patients with heteroplasmy (67% to 98%) elected to undergo at least one pronuclear-transfer procedure. Analysis of 67 enucleated zygotes from 10 of these patients showed that the majority of the patients (70%) had no zygotes with less than 30% heteroplasmy. Three of the women each had a single zygote with less than 30% heteroplasmy (range, 12 to 25) (Fig. S10). Overall, 4.5% of the enucleated zygotes had less than 30% heteroplasmy and 67.2% had more than 60%, a level that is associated with an increased risk of the development of disease symptoms.[9] Thus, although the possibility of obtaining zygotes with low heteroplasmy from patients undergoing pronuclear transfer is difficult to rule out, those who underwent pronuclear transfer most likely would not have benefited from PGT.

DISCUSSION

Building on our previous preclinical findings,[11] we found in the current study that clinical translation of pronuclear transfer was compatible with human embryo viability and reduced transmission of pathogenic mtDNA variants; analyses of neonatal blood showed a reduction in levels of maternal pathogenic mtDNA variants by 95 to 100% in six newborns and by 77 to 88% in two others. These data indicate that pronuclear transfer was effective in reducing transmission of mtDNA disease.

Given the potential for the amplification of the small fraction of maternal mtDNA that is cotransferred with the nuclear genome,[11,14-16] monitoring outcomes in a larger series will be essential to determine whether heteroplasmy remains stable over time and across different tissue types. For two babies born after pronuclear transfer, we detected 12% and 16% heteroplasmy for the maternal pathogenic variant. Although these levels are considered to be below the critical threshold for disease symptoms,[9] understanding the technical and biologic basis for these higher levels of variant heteroplasmy as compared with the other cases reported here will be important. A comparison of patient cytoplasm carryover scores indicated no marked difference among embryos with varying levels of heteroplasmy in the amount of cytoplasm surrounding the pronuclei. Potential explanations include a replicative advantage of patient mtDNA over the egg donor mtDNA[15,17] or unequal distribution of mitochondrial donation–induced heteroplasmy,[11,21-23] which would result in enrichment in a subset of embryonic cells that subsequently segregate to the epiblast lineage.[24]

Mitochondrial donation involves creating new combinations of mitochondrial and nuclear genomes. Studies involving the substitution of mouse mtDNA by backcrossing the nuclear genome of one inbred strain onto the cytoplasm of another[25-28] have shown a range of adverse effects, including on physical performance[29] and cognition,[27] whereas other studies have shown beneficial effects on metabolism and longevity.[28] However, the clinical relevance of these studies is unclear because human reproduction frequently involves combining nuclear and mitochondrial genomes from diverse ancestries, with no evidence of cotransmission of mtDNA or nuclear-encoded mitochondrial genes.[30,31] Moreover, population studies have shown that the coexistence of nuclear and mitochondrial genomes from diverse ancestries does not negatively affect human health.[32,33]

In accordance with the principle of cautious clinical application,[7] pronuclear transfer is offered only when PGT is unlikely to be successful. This series of 39 cases of PGT that resulted in 18 live births of infants with heteroplasmy levels ranging from undetectable to 7% (on the basis of 10 babies with known levels) adds to the existing body of evidence[34-39] and confirms the reliability of PGT in reducing the risk of mtDNA disease.

Performing a biopsy of trophectoderm cells at the blastocyst stage rather than on day 3 after intracytoplasmic sperm injection may further improve PGT outcomes.[40] However, reports of a clinically relevant discrepancy in levels of heteroplasmy[41] between biopsied trophectoderm cells and the corresponding infant[42] highlight the im-

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portance of further investigation to test the reliability of trophectoderm biopsy across a range of heteroplasmy levels.[5,6]

Although the incidence of clinical pregnancy was similar among patients in the two groups, pronuclear transfer was less efficient than PGT in establishing clinical pregnancies owing to an increased incidence of embryo loss after a positive pregnancy test. The success of pronuclear transfer was also curtailed by a lower percentage of normally fertilized eggs in the pronuclear-transfer group than in the PGT group. Vitrification of patients' eggs (which was carried out in most pronuclear-transfer procedures) may contribute to a lower incidence of fertilization; however, the effect varied among mtDNA variant types. This raises the possibility that the ability of an egg to undergo normal fertilization is compromised by the extent of biochemical defect caused by the variant it carries. Determining whether outcomes can be improved by performing spindle transfer, which would enable fertilization to occur in a wild-type cytoplasmic environment, would be of interest.

Our results show that a program involving PGT and pronuclear transfer was effective in reducing transmission of a range of pathogenic mtDNA variants. The reduced levels of heteroplasmy in infants born to women carrying homoplasmic variants provides grounds for optimism. However, until more is known about its efficacy, mitochondrial donation should be regarded as a risk-reduction strategy. In addition, clinical follow-up of children born after mitochondrial donation will be essential for monitoring the safety and efficacy of this procedure.

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